Statistics Mastery: Cheat-Sheet Pack

One reference table per course page, in roadmap order (36 items, 7 stages).

STAGE 1

item 1 Random variables, expectation, variance & covariance

Concept	Formula / one-liner
Random variable	A function X: outcomes → numbers
CDF	$F(x) = P(X \le x)$; $P(a < X \le b) = F(b) - F(a)$
Expectation	$E[X] = \Sigma\, x\, p(x) = \int x\, f(x)\, dx$; linear always: $E[aX+bY+c] = aE[X] + bE[Y] + c$
Indicator trick	$E[1_A] = P(A)$; expected count = sum of probabilities
Variance	$E[X^2] - (E\,X)^2$; $Var(aX+b) = a^2 Var(X)$
Covariance	$E[XY] - E[X]E[Y]$; $Corr = Cov/(\sigma_x \sigma_y) \in [-1,1]$ (linear only)
Var of a sum	$Var\,X + Var\,Y + 2Cov(X,Y)$; $w^T \Sigma w$ in general
Independent ⇒ ?	$Cov = 0$, $Var(X \pm Y) = VarX + VarY$; but $Cov = 0 \not\Rightarrow$ independent
Total expectation / variance	$E[Y] = E[E[Y\|X]]$; $Var\,Y = E[Var(Y\|X)] + Var(E[Y\|X])$
Standard error of the mean	$\sigma/\sqrt{n}$
Unbiased variance	s^2 with divisor $n - 1$ (s itself still biased)
Chebyshev / Markov	$P(\|X-\mu\| \ge k\sigma) \le 1/k^2$; $P(X \ge t) \le E[X]/t$
Jensen	convex g: $E[g(X)] \ge g(E[X])$
Mahalanobis	$d^2 = (x-\mu)^T \Sigma^{-1} (x-\mu) \sim \chi^2_p$ for normal data
Correlated Bernoulli sum	$Var(D) = nPD(1-PD)[1 + (n-1)\rho]$
CLT	$(\bar{X}-\mu)/(\sigma/\sqrt{n}) \to N(0,1)$; slow for skewed data and rare events

item 2 Core distributions and how they relate

Distribution	One line to remember
Binomial(n,p)	Successes in n independent trials; mean np, var np(1−p); normal-ish if np and n(1−p) ≥ 10
Poisson(λ)	Rare events in a window; mean = var = λ; limit of Binomial; sums of Poissons are Poisson
Exponential(λ)	Gap between Poisson events; memoryless; constant hazard; mean 1/λ
Gamma(a, β)	Sum of a exponentials; mean a/β; CV = 1/√a; $\chi^2(k)$ = Gamma(k/2, ½)
Normal	Sum of many small effects; closed under linear combinations; 95% within 1.96σ
Lognormal	exp of a normal; product of many factors; mean e^(μ+σ²/2) > median e^μ
$\chi^2(k)$	$\Sigma\, Z^2$; mean k, var 2k; $(n-1)s^2/\sigma^2 \sim \chi^2(n-1)$
t(ν)	$Z/\sqrt{(\chi^2/\nu)}$; fat tails; moments exist only below ν; → Normal as ν → ∞
$F(d_1,d_2)$	Ratio of scaled χ^2s; compares variances; $t^2 = F(1,\nu)$
Beta(a,b)	Distribution on [0,1]; posterior Beta(a+k, b+n−k); pseudo-counts

item 3 Conditional probability and Bayes' theorem

Idea	One line
Conditional probability	$P(A\|B) = P(A \cap B)/P(B)$: restrict to B, renormalise
Chain rule	$P(A \cap B \cap C) = P(A) \cdot P(B\|A) \cdot P(C\|A \cap B)$
Total probability	$P(B) = \Sigma\, P(B\|A_i) P(A_i)$ over a partition
Independence	$P(A \cap B) = P(A)P(B)$; lift = 1
Bayes	$P(A\|B) = P(B\|A)P(A)/P(B)$
Odds form	posterior odds = prior odds × LR; LR^+ = sens/FPR; LR^- = (1−sens)/(1−FPR)
Precision	πTPR / ($\pi TPR + (1-\pi)FPR$): driven by prevalence π as much as by the model
Decision threshold	Act when P(event | data) > c_{FP} / (c_{FP} + c_{FN})
Beta–Binomial	Beta(a,b) + k of n → Beta(a+k, b+n−k)
Gamma–Poisson	Gamma(a,b) + k events in n periods → Gamma(a+k, b+n)
Normal–Normal	Precisions add; posterior mean is precision-weighted
Shrinkage	(a+k)/(a+b+n): small segments pulled toward the book mean
Prosecutor's fallacy	P(match|innocent) ≠ P(innocent|match); expect (N−1)/X other matches
Credible vs confidence interval	Credible: probability statement about θ. Confidence: statement about the procedure.

item 4 Law of large numbers and the central limit theorem

Idea	One line
LLN	$\bar{X}_n \to \mu$ if $E\|X\| < \infty$ (weak: in probability; strong: almost surely)
Gambler's fallacy	No compensation: surplus stays ~√n, proportion converges by dilution
Chebyshev	$P(\|\bar{X} - \mu\| \geq \varepsilon) \leq \sigma^2/(n\varepsilon^2)$: always true, usually loose
Standard error	σ/√n: 4× data halves it; 100× data gains a digit
Required n	$n \approx (1.96\sigma/m)^2$ for margin m
CLT	$(\bar{X} - \mu)/(\sigma/\sqrt{n}) \to N(0,1)$; needs finite variance
Speed	$\text{skew}(\bar{X}) = \text{skew}(X)/\sqrt{n}$; Berry–Esseen error ≤ $0.4748 \cdot E\|X-\mu\|^3/(\sigma^3\sqrt{n})$
Heavy tails	α ≤ 1: no LLN; 1 < α ≤ 2: LLN but no √n CLT; Cauchy mean is Cauchy
Dependence	$n_{eff} = n/[1 + 2\Sigma\rho_k]$; AR(1): $n(1-\rho)/(1+\rho)$
Binomial → normal	N(np, npq) with continuity correction k ± ½; use Poisson for rare events
Delta method	$SE[g(\bar{X})] \approx \|g'(\mu)\|\sigma/\sqrt{n}$; breaks when $g'(\mu) = 0$
Slutsky	Replace σ by s and the CLT still holds ($s \to \sigma$)
Rare proportions	Precision is set by events: ±10% needs about 384(1−p) events; prefer Wilson
Correlated portfolio	Loss-rate spread does not vanish as N grows; independent-normal understates tails

STAGE 2

item 5 Sampling distributions, standard errors and estimators

Idea	One line
Sampling distribution	Distribution of a statistic over repeated samples; you see one draw
SD vs SE	SD: spread of data (σ). SE: spread of an estimate (σ/√n for a mean)
FPC	× √((N−n)/(N−1)) when sampling without replacement from a finite population
Common SEs	$\sqrt{(pq/n)}$; $\sqrt{(\sigma_1^2/n_1+\sigma_2^2/n_2)}$; median $1.2533\sigma/\sqrt{n}$; $(1-\rho^2)/\sqrt{n}$
MSE	Bias^2 + Variance
Variance divisors	n−1 unbiased; n+1 minimises MSE for normal data
Jensen	Unbiasedness does not survive nonlinear maps (s is low for σ, $1/\bar{X}$ is high for λ)
UMVU	Best unbiased estimator uses the sufficient statistic (German tank: (1+1/n)max − 1)
Efficiency	Relative to the data: mean for normal, median/trimmed for heavy tails; CRLB gives the floor
Shrinkage	$c^* = \mu^2/(\mu^2 + \sigma^2/n)$: trade a little bias for a lot of variance
Normal sampling	$\bar{X} \perp s$; $(n-1)s^2/\sigma^2 \sim \chi^2_{n-1}$; $t = (\bar{X}-\mu)/(s/\sqrt{n}) \sim t_{n-1}$
Bootstrap	SD of statistic over resamples with replacement; jackknife leaves one out
Design effect	DEFF = 1 + (m−1)ρ; n_{eff} = rows/DEFF; use cluster-robust SEs

item 6 Maximum likelihood estimation

Idea	One line
Likelihood	$L(\theta) = P(\text{data} \mid \theta)$, as a function of θ; not a distribution over θ
Score and MLE	$S(\theta) = \ell'(\theta)$; the MLE solves $S(\hat{\theta}) = 0$
Fisher information	$I(\theta) = -E[\ell'']$ per observation; $SE \approx 1/\sqrt{(nI(\hat{\theta}))}$
Closed-form MLEs	Bernoulli k/n; Poisson $\bar{x}$; exponential $1/\bar{x}$; normal $\mu = \bar{x}$, $\sigma^2 = SS/n$
Invariance	MLE of $g(\theta)$ is $g(\hat{\theta})$
Large-n properties	Consistent, asymptotically normal and efficient; small-sample bias O(1/n)
Wald / LR / score	LR and score are better near boundaries and are invariant (Wald is not)
Wilks	$2[\ell(\hat{\theta}) - \ell(\theta_0)] \to \chi^2_q$; 3.84 for one restriction at 5%
Newton	$\theta \leftarrow \theta - S/\ell''$; needs a good start; may find local maxima or leave the parameter space
Logistic regression	Concave log-likelihood, fitted by Newton/IRLS; $Cov = (X'WX)^{-1}$ with W = p(1−p)
Separation	MLE at infinity; use penalty (ridge/MAP) or Firth
Sandwich	$A^{-1}BA^{-1}$ gives valid SEs under misspecification; Poisson with overdispersion φ has SE too small by √φ
Censoring	Events contribute f(t), censored contribute S(c); rate = events/exposure
AIC	$-2\hat{\ell} + 2k$: likelihood penalised for the number of parameters

item 7 Confidence intervals

Idea	One line
Meaning	The procedure covers the truth in 95% of repeated samples; this one interval either does or does not
Mean	$\bar{x} \pm t_{n-1} \cdot s/\sqrt{n}$; using z with estimated σ undercovers at small n
Duality	The 95% interval = all θ_0 not rejected by the 5% test
Planning	$n = (z\sigma/m)^2$; proportion $z^2p(1-p)/m^2$; halve margin ⇒ 4× n
Proportion	Prefer Wilson, Jeffreys or likelihood-ratio; Clopper–Pearson if a guarantee is needed; avoid Wald
Zero events	Rule of three: upper 95% bound ≈ 3/n
Poisson / correlation / SD	Garwood (χ^2); Fisher z (atanh, SE $1/\sqrt{(n-3)}$); χ^2 interval for σ
Profile likelihood	Drop in the profile log-likelihood ≤ 1.92; asymmetric, invariant
Bootstrap	Percentile, basic, normal; BCa or studentised for accuracy; still approximate at small n
Comparing groups	Interval for the difference, not overlap of intervals (equal SE: significant beyond 2.77 SE, overlap until 3.92 SE)
Many intervals	P(all cover) = 0.95^m; Bonferroni uses level $1 - \alpha/m$
CI vs PI	CI width → 0 as n → ∞; PI width → ±1.96σ

item 8 Bootstrap and resampling

Idea	One line
Bootstrap	Resample n points with replacement from the data; recompute the statistic B times; its spread estimates the sampling spread
Centre	The bootstrap distribution is centred at $\hat{\theta}$, not θ: it estimates spread and shape, not location
SE / bias	SE = SD(θ*); bias = mean(θ*) − $\hat{\theta}$; bias-correct only when bias ≳ 0.25 SE
Jackknife	Leave-one-out; fine for smooth statistics, fails for the median
Failures	Maxima, infinite-variance tails, boundaries, tiny n, dependent data; use m-out-of-n, subsampling or a parametric bootstrap
Intervals	BCa or bootstrap-t for second-order accuracy; percentile/basic/normal are first-order
Regression	Pairs or wild bootstrap under heteroskedasticity; residual bootstrap needs constant variance
Permutation test	Shuffle labels under H_0; p = (1 + #extreme)/(1 + B); exact for any statistic if labels are exchangeable
Unequal variances	Raw permutation of the mean difference is invalid for equal-means H_0; studentise or use a bootstrap test that imposes H_0
Monte Carlo p	SE ≈ $\sqrt{(p(1-p)/B)}$; use B ≥ 999–9,999; never report p = 0
Time series	Moving block bootstrap, ℓ ≈ n^{1/3}; iid bootstrap understates SE by ≈ $\sqrt{((1+\varphi)/(1-\varphi))}$
Clusters	Resample whole clusters; design effect 1 + (k − 1)ρ
Model metrics	Stratified bootstrap for one AUC; paired (same cases) bootstrap for AUC differences

item 9 P-values from scratch to expert

Concept	One line
p-value	P(statistic at least this extreme | H_0 and all model assumptions true)
Under H_0	p ~ Uniform(0,1) (continuous statistics); P(p ≤ α) = α
Two-sided	2 × min(tail); decide direction before seeing data
CI duality	95% CI excludes null value ⇔ two-sided p < 0.05
z ≈ effect / SE	Large n shrinks SE → tiny effects become significant
Power	1 − β; ∝ effect·$\sqrt{n}/\sigma$; n ≈ $15.7/d^2$ per group for 80% / α=.05
m tests	P(≥1 false positive) = $1 - (1-\alpha)^m$; Bonferroni α/m; Holm; BH for FDR
False-positive risk	$\alpha(1-\pi) / [\alpha(1-\pi) + \text{power} \cdot \pi]$
Min Bayes factor	$BF_{01} \geq -e \cdot p \cdot \ln p$; p = .05 → ≥ 0.41
Surprisal	$s = -\log_2 p$ bits (p=.05 → 4.3 bits ≈ 4 heads in a row)
Permutation p	(count + 1) / (B + 1)
Equivalence	TOST: both one-sided tests reject ⇔ 90% CI inside ±Δ
Fisher combine	$-2 \Sigma \ln p_i \sim \chi^2_{2k}$
Never	say "probability H_0 is true", "accept H_0", or report only "p < 0.05"

items 9–10 Hypothesis testing, power and experimental design

Idea	One line
Errors	α = P(reject | H_0); power = P(reject | H_1) = 1 − β
Noncentrality	λ = effect / SE; power is the area of N(λ,1) beyond the critical value
MDE	≈ 2.8 SE for 80% power at two-sided α = 0.05 ($z_{1-\alpha/2}$ + z_power)
Sample size	$n = 2(z + z)^2/d^2$ per group; halve the effect ⇒ 4× n; 90% instead of 80% power ⇒ ×1.34
Effect sizes	Report d, risk difference, OR/RR, NNT, AUC = $\Phi(d/\sqrt{2})$; always with an interval
Observed power	A function of p; p = 0.05 ⇒ ~50%; adds nothing. Use the CI and pre-specified MDE instead
Type S / M	Low power + significance filter ⇒ exaggerated, occasionally wrong-signed estimates
CUPED	Variance × $(1 - \rho^2)$; covariate must be pre-treatment
Allocation	$n_1:n_2 = (\sigma_1/\sigma_2) \cdot \sqrt{(c_2/c_1)}$; 50/50 for equal variance and cost
Clusters	DE = 1 + (m − 1)ρ; power limited by number of clusters; plateau as m grows
Sequential	Peeking inflates α (5 looks ≈ 14%); Pocock = constant bound, O'Brien–Fleming = strict early, ~1.98 at the end
Heavy tails	Raw-mean tests lose power; log, cap or rank change the estimand; plan power by simulation

item 11 Linear regression done properly

Idea	One line
OLS	$\hat{\beta} = (X^TX)^{-1}X^Ty$; slope = Cov(x,y)/Var(x); residuals $\perp$ columns of X
FWL	A coefficient is the slope of residualised y on residualised x
Omitted variable	Bias of the short regression = $\beta_2 \cdot \delta$; control for confounders, not mediators or colliders
Precision	$SE(\hat{\beta_1}) = \sigma/(s_x\sqrt{(n-1)})$: more n, less noise, more spread in x
Assumptions	Linearity/exogeneity ⇒ unbiased; constant variance and independence ⇒ correct SEs; normality ⇒ exact small-n tests
R^2	Rises with every added predictor; use adjusted R^2 or out-of-sample R^2
Intervals	CI for the mean: $\sqrt{h_0}$; PI for a new point: $\sqrt{(1 + h_0)}$; both widen away from $\bar{x}$
Diagnostics	Residuals vs fitted (curve, funnel), Q–Q (tails), leverage and Cook's D (influence)
Influence	Needs high leverage and a sizeable residual; Cook's D > 4/n looks, > 1 alarms
Robust SE	HC3 for small n; WLS if variance is modelled; cluster-robust for grouped data
Measurement error	Slope on a noisy x shrinks by the reliability λ
Regression to the mean	Units selected for an extreme first value regress toward the mean by (1 − ρ) with no intervention
Log models	log y ⇒ ≈ 100β % per unit; exp(prediction) is the median: multiply by the smearing factor for means

item 12 Multicollinearity, interactions and nonlinearity

Idea	One line
Collinearity	Inflates variance by $1/(1 - R^2_j)$ (VIF), not bias; leaves predictions in-sample and joint effects intact
VIF	Regress x_j on the others; VIF = $1/(1 - R^2)$; $\sqrt{VIF}$ scales the SE; 10 is the old alarm level
Perfect collinearity	X^TX singular; coefficients unidentified (dummy trap)
Remedies	Joint F-test, combine variables, more data or design, ridge/PCA for prediction; no stepwise deletion
Interaction	Effect of x depends on d: β_1 is the slope at d = 0, β_3 the slope difference
Centring	Makes main effects mean "at the average"; same fit, better-behaved coefficients
Marginal effect	$\beta_1 + \beta_3 z$ with $Var = V_{11} + z^2V_{33} + 2zV_{13}$; plot it with a band
Interaction power	SE ≈ 2× that of a main effect; need ≈ 4× the sample for the same effect size
Polynomials	Global, wild at the edges and outside the data; rarely above degree 3
Splines	Local pieces joined at knots; hinge (linear) or truncated cubic basis; knots are the tuning parameter
Cross-validation	Choose flexibility by held-out error; one-SE rule favours the smoother model
Partial residuals	$e + \hat{\beta_j}x_j$ against x_j shows the shape of the effect of x_j
Binning	Median split loses ≈ 36% of the information; prefer splines or the raw variable

item 13 Generalised linear models

Idea	One line
GLM	$y \sim$ exponential family, $g(E[y]) = X\beta$; fit by IRLS (Newton for canonical links)
Logit	$\log(p/(1-p)) = X\beta$; $e^{\hat{\beta}}$ = odds ratio (constant); $\Delta p \approx \beta\, p(1-p)$ (varies)
OR vs RR	OR ≈ RR only when the event is rare (< ~10%)
Tests	Wald (SE), LR (two fits, best), score (null fit only); all χ^2_1
Separation	Perfect split ⇒ $\beta \to \infty$; use Firth or a penalty; MLE is biased outward in small samples
Metrics	AUC (rank), precision depends on base rate, calibration is separate from discrimination
Poisson	log link, $e^{\hat{\beta}}$ = rate ratio, offset log(exposure) with coefficient 1
Overdispersion	$\hat{\varphi}$ = Pearson/df > 1: quasi-Poisson, sandwich or negative binomial ($\mathrm{Var} = \mu + \alpha\mu^2$)
Deviance	$D = 2(\ell_{sat} - \hat{\ell})$; drop in D between nested models ~ $\chi^2(df)$; AIC = $-2\ell + 2p$
Pseudo-R^2	McFadden $1 - \hat{\ell}/\hat{\ell}_0$; 0.2–0.4 is already very good
Links	Logit ≈ probit (÷1.7); cloglog is asymmetric; they differ in the tails
Undersampling	Slopes unchanged, intercept pushed up by $-\log r$; add $\log r$ back before using probabilities

item 14 Regularisation: ridge, lasso and elastic net

Idea	One line
Why penalise	OLS variance ≈ $\sigma^2 p/n$; a little bias buys a large variance reduction
Ridge	$\hat{\beta} = (X^TX + \lambda I)^{-1}X^Ty$; shrinks along PCs by $d^2/(d^2+\lambda)$; never exactly zero; df = $\Sigma d^2/(d^2+\lambda)$
Standardise	Always scale predictors before penalising; leave the intercept unpenalised; do it inside CV folds
Lasso	L1 penalty ⇒ exact zeros; orthonormal case = soft-threshold $\mathrm{sign}(z)(\lvert z\rvert - \lambda)_+$
Algorithm	Coordinate descent with warm starts along a decreasing λ path; $\lambda_{max} = \max\lvert x_j^Ty\rvert/n$
Elastic net	$\alpha\lVert\beta\rVert_1 + (1-\alpha)\tfrac{1}{2}\lVert\beta\rVert^2$; grouping effect for correlated predictors; can select > n
Choosing λ	K-fold CV: λ_min (best) or λ_1se (sparser); λ itself is noisy
Inference	Coefficients are biased; post-selection p-values and CIs are invalid; use splitting or debiasing
Penalised GLMs	Penalised IRLS; cures separation; choose λ by CV log-loss or deviance
Adaptive lasso	Weights $1/\lvert\hat{\beta}^{\dagger}\rvert^{\gamma}$ give the oracle property; lasso alone over-selects
Stability selection	Selection frequency across half-samples; keep variables above ~60–80%
Bayesian view	Ridge = Gaussian prior, lasso = Laplace prior, penalised estimate = posterior mode

item 15 Model selection

Idea	One line
Error decomposition	$E[\text{error}] = \sigma^2 + \text{bias}^2 + \text{variance}$; U-shaped in flexibility
Optimism	Test error ≈ training error + $2k\sigma^2/n$ (Cp); training error always prefers the biggest model
AIC	$-2\ell + 2k$; aims at prediction (efficient); AICc for small n/k
BIC	$-2\ell + k \log n$; approximates marginal likelihood; consistent; prefers smaller models
Akaike weights	$w \propto \exp(-\Delta/2)$; Δ < 2 means essentially tied; average predictions across models
K-fold CV	K = 5–10; LOO nearly unbiased but high variance; LOO shortcut $e/(1-h)$ for linear models
Leakage	Repeat every data-dependent step (screening, scaling, imputation) inside each fold
Winner's curse	Best-of-M CV score is optimistic; use a test set or nested CV for the whole procedure
Stepwise	Invalid p-values, inflated R^2, unstable; prefer shrinkage or pre-specification
Time-ordered data	Out-of-time and rolling-origin validation; random CV leaks neighbours
Metrics	AUC for ranking; log-loss/Brier for probabilities; match the decision
Instability	Bagging, model averaging or shrinkage reduce the variance of selection

STAGE 4

item 16 Calibration and proper scoring rules

Idea	One line
Calibration	P(Y = 1 | forecast q) = q; separate from discrimination (AUC)
Reliability diagram	Binned mean forecast vs observed rate; below the diagonal = over-prediction; flat curve = over-confident
Proper rule	Expected loss minimised by reporting the true p; Brier, log-loss, spherical are strictly proper; accuracy and $\|q - y\|$ are not
Brier decomposition	BS = REL − RES + UNC; BSS = 1 − BS/UNC
Calibration slope / intercept	logit P(Y) = α + β logit q; ideal 0 and 1; β < 1 over-fit, β > 1 under-confident
ECE	Bin-weighted $\|f - o\|$; depends on bins and is upward-biased in small samples
Causes	Overfitting (slope < 1), shrinkage (slope > 1), resampling or class weights (intercept), model family, dataset shift
Recalibration	Platt / temperature (logistic), isotonic (monotone steps, needs data), fit on separate data
Base-rate shift	Add $\log[(\pi'/(1-\pi'))/(\pi/(1-\pi))]$ to the logit when only the prior moved
Log-loss vs Brier	Log-loss punishes confident errors without limit; Brier is bounded; AUC cannot see calibration
Segments	Average calibration can hide opposite errors; check by segment
Decisions	t* = c_FP/(c_FP + c_FN) needs calibrated q; NB(t) = TP/n − FP/n · t/(1 − t)
Tests	HL, Spiegelhalter, binomial by grade: reject too often at large n, too rarely at small n; report slope with CI

item 17 Imbalanced data and rare events

Idea	One line
Accuracy paradox	Predicting "no event" is 99.5% accurate at 0.5% prevalence; use recall, precision, PR curves, MCC
ROC vs PR	ROC-AUC is prevalence-free; precision falls with prevalence; PR baseline = prevalence
PPV	$\text{sens}\cdot\pi / [\text{sens}\cdot\pi + (1 - \text{spec})(1 - \pi)]$; tiny π means most alerts are false
Operating point	F1 (threshold ≈ F*/2 for calibrated p), cost t* = c_FP/(c_FP + c_FN), or capacity top-m
Information	Comes from events; ≥ 10 events per parameter is a minimum, check Riley-type criteria
Resampling	No new information; shifts probabilities; do only inside training folds
Prior correction	Undersample non-events with fraction r: add ln r to the intercept; slope variance × ≈ (1 + π/r)
Rare-event bias	ML under-predicts event probability, can separate; use Firth, ridge or Bayesian priors
Cost-sensitive	Weight w on events ≡ threshold 1/(1 + w) for a correct model; weighting costs variance and calibration
Evaluation	Stratify, repeat, report intervals; out-of-time test; don't tune the threshold on the test set
Rare rates	Wald fails; Clopper–Pearson (conservative), Wilson, Jeffreys; 0 events → upper bound ≈ 3/n

item 18 Survival and time-to-event analysis

Idea	One line
Censoring	Right-censored: we know $T > c$; assume non-informative; never drop or recode as non-events
S, h, H	$S = \exp(-H)$; $h = f/S$; mean $= \int S$; $\text{RMST}(\tau) = \int_0^\tau S$
Kaplan–Meier	$\hat{S} = \prod(1 - d/n)$; Greenwood variance; log-log CI; show numbers at risk
Log-rank	Observed − expected events by group; best under PH; weights $G(\rho, \gamma)$ for early/late effects; crossing curves need other tests or RMST
Cox	$h = h_0(t)\exp(x\beta)$; partial likelihood; $\text{HR} = \exp(\beta)$; information is the number of events
Prediction	$\hat{H}_0$ by Breslow; $\hat{S}(t\|x) = \exp(-\hat{H}_0 e^{x\beta})$; HR ≠ absolute risk; C-index for ranking
PH check	Schoenfeld residuals, log-log plots, time interactions; if violated the HR is a follow-up-dependent average
Competing risks	1 − KM overstates; use cumulative incidence (Aalen–Johansen); cause-specific vs Fine–Gray
Discrete time	Person-period logistic hazards; $\hat{S} = \prod(1 - \hat{h})$; time-varying covariates; lifetime PD term structure
Evaluation	Horizon AUC, Brier and calibration with IPCW or KM; Harrell's C depends on censoring
Parametric	Likelihood: events log f, censored log S; AFT time ratios; extrapolation is assumption-driven

item 19 Missing data

Idea	One line
MCAR / MAR / MNAR	Missingness independent of data / depends on observed data / depends on the missing value itself; an assumption, not testable from the data alone
Complete cases	Unbiased for regression if missingness depends on covariates, not the outcome; biased if it depends on the outcome; always loses precision
Single imputation	Mean fill shrinks variance and correlation; regression fill inflates correlation; even stochastic fill gives SEs that are too small
Multiple imputation	Draw M completions from the posterior predictive (parameters too), analyse each, pool with Rubin: $T = \bar{W} + (1 + 1/M)B$
Fraction missing info	$\gamma = (1 + 1/M)B/T$; efficiency $(1 + \gamma/M)^{-1}$; use M of at least the percentage of incomplete cases
Imputation model	Include the outcome, all analysis variables (and interactions) and predictors of missingness; chained equations for many incomplete variables; check convergence
Skewed variables	Transform or use predictive mean matching, which imputes real observed donor values
MNAR	Not identifiable; do delta-adjusted or reference-based sensitivity analysis and report the tipping point
Weighting	IPW $= 1/P(\text{observed} \mid x)$, needs positivity; AIPW is doubly robust (either the outcome or the selection model right)
Reject inference	Outcome missing for rejected applicants; MAR if the policy used recorded variables; manual overrides make it MNAR
Prediction models	Imputation rule fixed on training data; indicator helps if missingness is informative but is fragile under deployment shift; monitor missing rates
Reporting	Amount and patterns, mechanism argument, imputation model, M and seed, pooled results with γ, sensitivity analysis

item 20 Dependent data: time series, panels, mixed models

Idea	One line
Effective sample size	Positive dependence repeats information: AR(1) $n_{eff} \approx n(1-\rho)/(1+\rho)$; clusters: $n/(1+(m-1)ICC)$
ACF / PACF	AR(p): ACF decays, PACF cuts off at p. MA(q): ACF cuts off at q, PACF decays. Bars within $\pm 1.96/\sqrt{n}$ are noise
Stationarity	Constant mean, variance, lag-covariance; AR roots outside the unit circle; random walk is not stationary
HAC	Newey–West with Bartlett weights and L growing with n; repairs the SE, not the efficiency or the model
Spurious regression	Independent random walks look related; $R^2 >$ Durbin–Watson is a warning; difference or test for cointegration
Cointegration	Integrated series with a stationary combination: use level relation plus error-correction; differencing loses it
Forecast evaluation	Walk-forward, never random k-fold; benchmarks (naive, seasonal naive, mean); error vs horizon; interval coverage
Clustered data	Design effect $1+(m-1)ICC$; cluster-robust SEs need many clusters; cluster where treatment or dependence lives
FE vs RE	FE: consistent, within variation only. RE: efficient but needs α_i uncorrelated with x. Hausman and Mundlak help choose
Mixed models	REML variance components; BLUP shrinks by $\lambda = \sigma^2_u/(\sigma^2_u + \sigma^2_e/n_j)$: partial pooling beats no and complete pooling
Binary clusters	Conditional (GLMM) ≠ marginal (GEE, pooled logit with robust SE); marginal $\approx \beta/\sqrt{1+0.346\sigma^2_u}$
GARCH / VaR	$\sigma^2_t = \omega + \alpha r^2_{t-1} + \beta\sigma^2_{t-1}$; backtest rate and independence; constant-σ VaR clusters its exceedances
Breaks	Ignored level shifts mimic unit roots; Chow (known date), sup-F/CUSUM (unknown), model or window the break

item 21 Multiple testing and selective inference

Idea	One line
Multiplicity	P(≥ 1 false positive) = 1 − (1 − α)^m; the family includes hidden choices (subgroups, outcomes, cut-offs, models)
FWER vs FDR	FWER = P(V ≥ 1), for confirmatory claims. FDR = E[V/R], for screening; neither is controlled by testing each at α
Bonferroni / Holm	α/m; Holm is step-down with α/(m − k + 1) and dominates Bonferroni under any dependence
Benjamini–Hochberg	Largest k with $p_{(k)} \le kq/m$; FDR = $\pi_0 q$ under independence or PRDS; BY divides q by Σ1/i for arbitrary dependence
Storey / q-value	$\hat{\pi_0}$ = #{p > λ}/((1 − λ)m); BH at $q/\hat{\pi_0}$; more power when many effects are real
FDR is an average	The FDP of one list varies, more so with correlated tests
Winner's curse	Selected estimates are biased up; confirm on new data, shrink (empirical Bayes), use hierarchical models
Subgroups	Test the interaction once; "significant here, not there" is not evidence of heterogeneity; pre-specify and adjust
Selective inference	Naive intervals fail after selection; use simultaneous, conditional (truncated), split or replication intervals
Backtest overfitting	Best of K noise strategies has Sharpe ≈ SE·√(2 ln K) in sample and 0 out of sample; deflate by the number of trials
Local fdr, empirical null	$fdr(z) = \pi_0 f_0/f$; estimate the null from the bulk when the theoretical N(0,1) is too narrow
Peeking	Repeated looks inflate α; use Pocock, O'Brien–Fleming or alpha-spending boundaries, fixed in advance
PPV	PPV = π·power/(π·power + (1 − π)α); low priors and flexibility make many significant results false; replicate

item 22 Confounding, selection bias, colliders, Simpson's paradox

Idea	One line
Confounder	Common cause of treatment and outcome; naive slope = $\beta + \kappa\gamma/(1 + \gamma^2)$ in the linear case; does not shrink with n
Simpson's paradox	A trend reverses on pooling; which table is right depends on the diagram (confounder: stratify; mediator or collider: do not)
Chain, fork, collider	Chain and fork: conditioning blocks. Collider: conditioning (or on its descendant) opens the path
Backdoor criterion	Adjust for a set with no descendant of X that blocks all backdoor paths; include confounders, exclude mediators and colliders
Precision variables	Causes of Y only help variance; causes of X only hurt it
Collider / selection bias	Filtering on a common effect makes causes dependent (Berkson); worst when selection is tight and driven by the variables studied
Over-adjustment	Do not control for post-treatment variables for the total effect; controlling for a mediator with an unmeasured mediator–outcome cause biases even a randomised effect
Reject inference	Selection on observables is harmless if the model is right, but extrapolation fails; selection on unobservables biases; only exploration is assumption-free
Proxies and M-bias	A noisy control removes only part of the confounding; adjusting for a pre-treatment collider creates bias
Regression to the mean	Expected change of the selected = −(1 − r) × their standardised time-1 mean; use a randomised control from the same selection
Sensitivity	OVB = $\kappa\gamma/(1 + \gamma^2)$; E-value = RR + √(RR(RR − 1)): the strength a hidden confounder needs with both X and Y
Negative controls	An outcome the treatment cannot affect should show no effect; a non-zero estimate exposes confounding; calibrate only under a defended proportionality

item 23 Potential outcomes, A/B testing, CUPED, sequential testing

Idea	One line
Potential outcomes	$\tau_i = Y_i(1) - Y_i(0)$ is never observed; naive difference = ATT + selection bias; randomisation sets the selection term to zero
SUTVA	No interference between units and a single version of treatment; fails with spillovers, shared capacity, networks
Analysis	Difference in means with the unpooled (Neyman) SE; randomisation test for exact, assumption-light inference
Sample size	n per arm = $2(z_{1-\alpha/2} + z_{1-\beta})^2\sigma^2/\delta^2$; halving the MDE costs 4× the sample; set the MDE from the decision
Clustering	DEFF = $1 + (m - 1)\rho$; analyse at the cluster level; more clusters beat more customers per cluster
Interference	Individual randomisation estimates a partial-exposure effect; randomise clusters or use switchbacks for the rollout effect
Regression adjustment	Adjust for pre-treatment covariates: unbiased, variance × $(1 - R^2)$; never adjust for post-treatment variables
CUPED	$\tilde{Y} = Y - \theta(X - \bar{X})$, $\theta = \mathrm{Cov}(X,Y)/\mathrm{Var}(X)$; X from the pre-period; variance × $(1 - \rho^2)$; longer pre-periods help
SRM and attrition	Chi-square on arm counts at $p < 0.001$; unequal loss re-introduces selection bias inside an experiment; analyse by assignment
Non-compliance	ITT for the offer; as-treated is biased; CACE = ITT / take-up difference is a local (complier) effect
Sequential testing	Peeking inflates α; use group-sequential boundaries or always-valid mSPRT with $\Lambda_n \geq 1/\alpha$; early-stopped estimates are biased up
Bandits	Thompson sampling lowers regret, sacrifices clean inference on weaker arms; use when the goal is reward, not estimation

item 24 Observational causal methods (matching, IPW, DiD, IV, RDD)

Idea	One line
Ignorability + overlap	$(Y(0),Y(1)) \perp T \mid X$ and $0 < e(x) < 1$; the propensity score e(X) is a balancing score; check SMDs and the overlap plot
Matching	Pair like with like (usually on the logit ps); caliper trades bias for a changed population; only observed confounders are handled
IPW, AIPW	Weights 1/e, 1/(1 − e); AIPW is consistent if either the propensity or the outcome model is right; add cross-fitting with ML
Overlap and trimming	Extreme weights ⇒ huge variance; trimming changes the estimand; effective sample size $(\Sigma w)^2/\Sigma w^2$
Difference-in-differences	Change in treated − change in control; needs parallel trends, no anticipation, no spillover; pre-trends are a check, not a proof
Staggered adoption	TWFE uses already-treated units as controls; with dynamic effects it can be biased or wrongly signed; use clean cohort-time estimators
Instrumental variables	$\beta = \mathrm{Cov}(Z,Y)/\mathrm{Cov}(Z,X)$; relevance, independence, exclusion; estimates the LATE for compliers
Weak instruments	$F < \sim 10$ ⇒ bias towards OLS and heavy tails; exclusion violation adds ζ/π; use Anderson–Rubin intervals
Regression discontinuity	Jump of local linear fits at the cut-off; continuity assumption; bandwidth = bias–variance; effect holds at the threshold
Fuzzy RD, manipulation	Wald ratio of the jumps for compliers; check density, covariates, donut, placebo cut-offs
Synthetic control	Convex-combination of donors matching the pre-path; placebo permutation inference; fails outside the hull
Method choice	Pick by what you can defend: randomisation > RD/IV/DiD with diagnostics > adjustment on observables; triangulate and report the assumption

item 25 Uplift and heterogeneous treatment effects

Idea	One line
CATE and uplift	$\tau(x) = E[Y(1) - Y(0) \mid X = x]$; uplift is the change in conversion probability caused by treatment
Four types	Persuadables (+), sure things (0, waste), lost causes (0), sleeping dogs (−); response models rank sure things first
Policy rule	Treat if $V \cdot \tau(x) > c$; profit curve peaks where the marginal uplift equals c/V
Subgroups	Difference in means within pre-specified subgroups; fine subgroups are noisy; shrink towards a common effect
Meta-learners	S: one model with T as a feature (under-detects). T: separate models (noisy, doubles variance). X: imputed effects weighted by propensity (good with unbalanced arms)
Causal trees and forests	Split on effect differences; adaptive leaf estimates are biased; honest splitting separates structure from estimation; forests give intervals
Qini / AUUC	Cumulative incremental gain by targeted share on a randomised test set; area against the random line; a response model scores near random
Winner's curse in targeting	Top-ranked customers' predicted uplift overstates the truth; validate with held-out randomised data or shrink
DR-learner / R-learner	Orthogonal pseudo-outcome ψ regressed on X; needs ignorability and overlap; cross-fit ML nuisances
Heterogeneity test	BLP on held-out data: coefficient on $T \cdot (\hat{\tau} - \text{mean})$ is 1 if calibrated, 0 if no signal; avoid hunting subgroups
Off-policy evaluation	Direct (model), IPW (unbiased, noisy), doubly robust; needs randomised or known logging with exploration
Pilot and holdout	Pilot size pays off with heterogeneity and margin; keep a permanent holdout and an exploration band
Fairness constraints	Parity of treatment rates costs value; measure the price of fairness; monitor by group; document the causal basis

item 26 Bayesian inference

Idea	One line
Bayes for parameters	$p(\theta \mid y) \propto p(y \mid \theta)\, p(\theta)$; the posterior is the prior reweighted by the likelihood
Beta–binomial	Beta(a, b) prior and k of n gives Beta(a + k, b + n − k); posterior mean is a weighted average with weight n/(n + a + b) on the data
Normal–normal	Precisions add; posterior mean is precision-weighted; the prior is worth σ^2/s_o^2 observations
Gamma–Poisson	Gamma(a, b) prior gives Gamma(a + Σy, b + n); predictive distribution is negative binomial
Sequential updating	Yesterday's posterior is today's prior; same result as one batch
Priors	Flat is not neutral; Jeffreys is reparametrisation-invariant; prefer weakly informative or documented informative priors; check by simulation and report sensitivity
Credible intervals	Equal-tailed or HPD; probability statement about θ given prior and data; coverage is exactly nominal on average over the prior only
Predictive checks	Simulate replicate data from the fitted model, compare a failure-prone statistic; the mean is usually uninformative
Bayesian regression	Normal prior gives the ridge estimate with $\lambda = \sigma^2/\tau^2$ plus a posterior covariance $\sigma^2(X'X + \lambda I)^{-1}$
Asymptotics	Bernstein–von Mises: the prior washes out unless it gives zero (or negligible) mass to the truth; the likelihood principle: the stopping rule does not change the posterior
Bayes factors	Ratio of marginal likelihoods; Occam effect; sensitive to the prior under the alternative; Jeffreys–Lindley: fixed p, growing n favours H_o
Empirical Bayes	Estimate the prior from many units, shrink each estimate in proportion to its noise; cures the winner's curse in rankings
Bayesian A/B	P(B > A), expected loss, ROPE; peeking at a threshold rule inflates false wins, so calibrate by simulation
Low-default PD	External beta prior plus own data; report prior source, strength, sensitivity and the data-only upper bound

item 27 MCMC and variational inference

Idea	One line
Why sample	Posterior summaries are integrals; Monte Carlo error is 1/√M in any dimension, whereas a grid needs 100^d points
Importance sampling	Weights w = p/q; ESS = $(\Sigma w)^2/\Sigma w^2$; the proposal needs tails at least as heavy as the target, or the weights have infinite variance
Metropolis–Hastings	Accept with $\min(1, p(\theta^*)/p(\theta))$: only a ratio, so the normalising constant cancels; step size tuned for about 44% (1-d) or 23% (many-d) acceptance
Gibbs	Draw each parameter from its full conditional; no tuning, always accepted; slow when parameters are highly correlated (ESS fraction $(1-\rho^2)/(1+\rho^2)$ for a bivariate normal)
Warm-up and chains	Several chains from dispersed starts; discard warm-up; fix and record the seed
$\hat{R}$	Between- over within-chain variance on split chains; require below 1.01; blind to modes no chain visits
ESS and MCSE	ESS = n/(1 + 2Σρ_k); MCSE = sd/√ESS; do not thin to gain precision; check bulk-ESS and tail-ESS
HMC and NUTS	Gradient-guided long moves with a Metropolis correction; ε near the narrowest width; NUTS picks the trajectory length; divergences signal biased exploration
Logistic regression	Standardise predictors; use the Laplace covariance to shape proposals; correlated predictors give correlated coefficients
Laplace approximation	Normal at the mode with the inverse Hessian; exact for normal posteriors; poor for skewed, bounded or multimodal ones
Variational inference	Maximise the ELBO (minimise $KL(q \,\|\, p)$) over a simple family; fast; mean-field underestimates variance and has thin tails
Tail probabilities	Relative error ≈ $\sqrt{((1-p)/(p \cdot ESS))}$: rare events need many effective draws
Workflow	Test on simulated data, several chains, $\hat{R}$ and ESS and divergences, MCSE below the posterior sd, predictive checks, prior sensitivity

item 28 Hierarchical (partial pooling) models

Idea	One line
Three pooling levels	No pooling: noisy, unbiased; complete pooling: stable, biased; partial pooling: estimates how different the groups are and weighs accordingly
Normal hierarchical model	$y_j \sim N(\theta_j, \sigma^2/n_j)$, $\theta_j \sim N(\mu, \tau^2)$; posterior mean $(1 - B_j)\bar{y}_j + B_j\mu$ with $B_j = (\sigma^2/n_j)/(\sigma^2/n_j + \tau^2)$
Who is shrunk most	Small groups and similar groups (small τ); the posterior variance $(1 - B_j)\sigma^2/n_j$ falls for every group
Estimating τ	Learned from J group means: wide posterior with few groups; point estimates often hit zero; use the marginal posterior of τ
Priors on τ	Weakly informative on the sd (half-Cauchy, half-normal, flat with enough groups); avoid inverse-gamma(ε, ε) on τ^2; check sensitivity
Funnel	Small τ forces the θ_j together: samplers struggle; non-centre with $\theta_j = \mu + \tau\eta_j$; divergences are the warning
Full vs empirical Bayes	Empirical Bayes plugs in estimated hyperparameters and gives intervals that are too short; full Bayes integrates them out
Varying slopes	Shrink the (intercept, slope) pair towards the population mean along their correlation; prior on Σ via scales and an LKJ correlation
Group predictors	$\theta_j = \gamma_0 + \gamma_1 z_j + u_j$: shrink towards a prediction, pay only for the residual spread; beware ecological fallacy
New groups	Predict from the population distribution; calibrated and narrower than own-data-only intervals; back-test on held-out groups
Rankings	Rank by posterior expected rank or probability of being in the worst k%; report rank intervals
When pooling hurts	Exchangeability fails for unusual groups: add predictors, use a t or mixture population, check with predictive checks and leave-one-group-out CV

item 29 Bayesian decision theory

Idea	One line
Bayes action	Choose the action with the smallest posterior expected loss, $E[L(a, \theta) \mid y]$
Loss and summary	Squared loss → mean; absolute loss → median; 0–1 loss → mode; linear asymmetric loss → quantile
Quantile rule	Under-cost c_u, over-cost c_o: act at the $c_u/(c_u + c_o)$ quantile of the predictive distribution
Threshold	Act if $p > c_FP/(c_FP + c_FN) = 1/(1 + c)$, for calibrated p; independent of the base rate
Calibration first	Recalibrate on held-out labels before applying a cost-based cut-off to a score
EVPI and EVSI	EVPI = expected loss of deciding now; EVSI(n) rises towards EVPI; collect data while EVSI − cost increases
Parameter uncertainty	Plug-in quantiles are too low; the posterior predictive (t for unknown mean and sd) is calibrated
Utility	Exponential utility, normal outcome: maximise mean − $(\lambda/2)\cdot$variance, including parameter uncertainty
Sequential stopping	Stop when posterior expected loss of the leader is below the cost of waiting; check error rates by simulation
Prior sensitivity	Map the decision over defensible priors; report whether it flips
Risk and minimax	Risk is MSE at each θ; Bayes risk averages it over the prior; minimax guards the worst case; shrinkage bets on the prior
Limited capacity	Rank by expected value per unit of capacity (p × exposure − cost); the oracle is the ceiling

item 30 Multivariate methods

Idea	One line
Covariance matrix	$\Sigma = V\Lambda V^T$: eigenvectors are the axes of the data cloud, eigenvalues the variances along them
Sample eigenvalues	The largest is over-estimated and the smallest under-estimated; relative sd about $\sqrt{(2/n)}$
Mahalanobis distance	$d^2 = (x - m)^T S^{-1}(x - m) \sim \chi^2_p$ for normal data; sees unusual combinations that Euclidean distance misses
PCA	Rotate to eigenvectors; keep k components; standardise first when units differ
How many components	Parallel analysis beats Kaiser, the 80% rule and the elbow; combine with interpretability and stability
PCA vs factor analysis	PCA explains total variance; FA models shared variance plus unique noise, $\Sigma = \Lambda\Lambda^T + \Psi$; PCA loadings are biased up with few variables
Rotation	Any orthogonal rotation fits equally well; varimax gives simple structure; oblique rotation for correlated factors
Mixtures and EM	E step: soft memberships; M step: weighted parameters; use several starts and BIC; check stability
High dimension	Sample covariance is unreliable when p/n is large; use shrinkage (Ledoit–Wolf) or structure
Robust covariance	Outliers mask themselves in classical Mahalanobis distance; MCD fits the tightest half
PCR trap	High variance does not mean predictive; choose k by cross-validation or use PLS or ridge
Distance in high dimension	Irrelevant variables dilute anomaly signals; curate features, reduce, shrink and test on planted anomalies
Stability	Axis error $\approx \sqrt{(\lambda_j\lambda_k/(n(\lambda_j-\lambda_k)^2))}$; small eigenvalue gaps make single components unstable; bootstrap them

item 31 Extreme value theory and heavy tails

Idea	One line
Heavy tail	$P(X > x) \approx x^{(-\alpha)}$: straight line on a log-log plot; the largest observation is a big share of the total
Moments	Finite only below α: $\alpha \le 1$ no mean, $\alpha \le 2$ no variance; LLN, CLT, correlation and regression then fail
Block maxima / GEV	Maxima converge to GEV(μ, σ, ξ); $\xi > 0$ heavy ($\alpha = 1/\xi$), $\xi = 0$ Gumbel, $\xi < 0$ bounded; wasteful and slow to converge
Peaks over threshold	Excesses over a high u follow GPD(σ, ξ); use mean excess and stability plots to choose u
Return level	$x_T = u + (\sigma/\xi)[(np/k)^{(-\xi)} - 1]$; extrapolation beyond the record is a model, not data; give intervals
Normal and empirical	A normal fit understates tail quantiles; an empirical quantile cannot exceed the sample maximum
Hill estimator	$\hat{\alpha}$ = 1/mean of log-ratios of the k largest; relative sd $1/\sqrt{k}$; bias grows with k; read the Hill plot
Clustering	Extremal index θ = 1/mean cluster size; decluster and count events; raw exceedances understate uncertainty
VaR and ES	$ES = VaR \cdot \alpha/(\alpha - 1)$ for Pareto, infinite if $\alpha \le 1$; ES is harder to estimate and biased low from samples; use a tail model
Diversification	One big jump: VaR of the average scales $m^{(1/\alpha - 1)}$; poor for α near 1, can reverse for $\alpha < 1$
Parameter uncertainty	Wald intervals undercover for small k; use profile likelihood or bootstrap; return-level intervals are wide and skewed
Operational losses	Compound Poisson-Pareto: $VaR_q \approx F^{-1}(1 - (1 - q)/\lambda) + (\lambda - 1)E[X]$; the tail index drives capital

item 32 Copulas and dependence modelling

Idea	One line
Why not correlation	Pearson depends on the margins and the bulk; the same ρ can hide very different joint tails; not defined without second moments
Sklar's theorem	H(x, y) = C(F(x), G(y)): model the margins and the copula separately; the copula is invariant to increasing transformations
Rank correlation	Kendall τ and Spearman ρ_s belong to the copula, need no moments and resist outliers; Gaussian ρ = sin(ττ/2)
Gaussian copula	Easy and scalable, but λ = 0 for ρ < 1: joint extremes fade; understates clustering of defaults and crashes
Archimedean	Clayton: lower-tail dependence 2^(−1/θ); Gumbel: upper-tail 2 − 2^(1/θ); Frank: none; exchangeable, one parameter
t copula	Symmetric tail dependence rising as ν falls and ρ rises; ν → ∞ gives the Gaussian
Tail dependence	λ = lim P(Y extreme | X extreme): a limit that is hard to estimate from few corner points
Portfolio risk	Same margins and τ, different copula: different VaR and ES of the sum, more so at high q; diversification benefit depends on C
Estimation	Pseudo-observations (ranks/(n + 1)) avoid margin misspecification; IFM needs correct margins; Kendall inversion is simple and robust
Model choice	Compare with the empirical copula (Cramér–von Mises, parametric bootstrap); low power in the tails; report sensitivity
Conditional correlation	Correlation in a subsample selected on a variable is biased (falls for a tail on X, rises for a volatile regime); use tail measures
Credit factor model	Vasicek quantile $\Phi((\Phi^{-1}(PD) + \sqrt{\rho}\,\Phi^{-1}(q))/\sqrt{(1-\rho)})$; capital driven by ρ; a t factor fattens the tail

item 33 Anomaly and drift detection theory

Idea	One line
Base rate	Precision = πTPR/(πTPR + (1 − π)FPR); rare anomalies make most alerts false even for a low false-alarm rate
Robust limits	Median ± 3 × 1.4826·MAD resists contamination (breakdown 50%); mean ± 3 sd is masked by the outliers it should find
Mahalanobis	$d^2 = (x-\mu)^T\Sigma^{-1}(x-\mu)$, χ^2_p under normality; finds combinations unusual across correlated variables; estimate Σ robustly
Isolation and density	Isolation forest: short average path = anomaly; k-NN and LOF are local density scores; no method wins on all shapes
Evaluation	AUC ignores prevalence; use precision–recall, average precision and precision at review capacity; labels are selected by the detector
Control charts	Shewhart for big jumps, EWMA and CUSUM for small shifts; quote ARL_0 and detection delay together
Many features	$1 - (1-\alpha)^m$ false alert chance; Bonferroni for any false alert, BH for the false discovery proportion; reduce the number of tests
Drift measures	PSI is biased up in small samples (≈ (B − 1)$(1/n_1 + 1/n_2)$); KS rejects trivial shifts at large n; use effect sizes with thresholds
Types of drift	Covariate: P(x) changes; concept: P(y|x) changes; prior: P(y) changes; label-free monitors cannot see concept drift
Window design	Short windows: fast, noisy; long windows: smooth, late; the operating curve is false alarms against delay
Cost-based thresholds	ARL_0 grows exponentially in h, delay linearly; best h rises with false-alarm cost and falls with drift frequency
Governance	Document the reference period, metrics, thresholds, costs, base rates, outcome lag and the response for each type of drift

item 34 Graph and network statistics

Idea	One line
Degree distribution	Random graphs: Poisson, no hubs; real networks: heavy tail from preferential attachment; the mean describes almost no node
Small world	High clustering with short paths; short paths are normal, so "within k hops" needs a reference
Centrality	Degree, closeness, betweenness (brokerage), eigenvector and PageRank answer different questions and can disagree strongly
Communities	Modularity is optimistic (random graphs score > 0); detectable only if $(a - b)^2 > 2(a + b)$; check stability
Null models	Erdős–Rényi is usually the wrong null; configuration model keeps degrees; p = (1 + #≥)/(1 + B)
Partial observation	Sampled networks give biased statistics (mean degree × f, clustering × f^2 or f^3); no general bootstrap
Neighbour features	Count and share of risky neighbours work only with homophily and known labels; beware leakage and hubs
Rings	Densest subgraph needs a null distribution of the maximum density; small loose rings cannot be told from chance
Homophily vs influence	Neighbour outcome regressions confound the two; designs and causal tools are needed; interference breaks SUTVA
Personalised PageRank	Risk propagation from seeds; damping sets reach; tune on later periods; exclude seeds in evaluation
Measurement error	All rankings degrade; hub-based measures hold up better than betweenness and paths; do a perturbation analysis and report stability
Governance	Document link definitions, coverage, null models, stability and false-positive rates of any network-based alert

item 35 Uncertainty quantification for ML

Idea	One line
Intervals	Confidence interval = where the mean is (width → 0); prediction interval = where a new case falls (width ≥ ±1.96σ)
Coverage	Marginal is easy; conditional on x is not; always report coverage by segment
Split conformal	$\hat{q} = \lceil (n + 1)(1 - \alpha) \rceil$-th smallest calibration score; coverage ≥ 1 − α on average; per calibration set ~ Beta(n + 1 − l, l)
Scores	Absolute, normalised, CQR change shape and efficiency, not validity; never tune on the calibration set
Classification	Sets from $1 - \hat{p}_y$; valid for any model; use class-conditional thresholds for rare classes
Covariate shift	Weight by p_{new}/p_{old}; watch $(\Sigma w)^2/\Sigma w^2$; concept drift is not repaired
Drift	ACI: $\alpha_{t+1} = \alpha_t + \gamma(\alpha - err_t)$; long-run coverage only; rolling coverage is a monitor
Ensembles	Spread = epistemic only; understates a new case's uncertainty; calibrate with conformal
Small groups	Per-group calibration needs $n_g > 1/\alpha$ and is noisy; pool, normalise or merge groups
Abstention	Abstain on wide intervals; value = gap to random abstention; monitor the decided subset
Model-based intervals	Valid only if the model is right; check actual against nominal on held-out data
Governance	Document the score, the calibration set, the exchangeability assumption, coverage by segment and drift monitoring

item 36 Interpretability and its statistical limits

Idea	One line
Importance	A property of the fitted model; unstable under correlated features; report variability over refits and group correlated features
Permutation importance	Reliance of the model; inflated for correlated features by off-distribution shuffles; conditional permutation and drop-column answer other questions
PD and ICE	The average hides interactions; plot ICE curves and the spread of individual effects; ALE for correlated features
Shapley values	Credits sum to prediction − baseline; interventional vs conditional differ; the baseline changes everything; not an effect
Sampling error	Importance estimates have intervals; do not rank features whose intervals overlap
LIME	Kernel width trades bias against variance; repeat and report variability; check local fidelity
Rashomon set	Many near-equal models, different stories; report what is common to the set
Causality	Explanations describe the model, not the effect of intervening; use causal methods for advice and policy
Counterfactuals	Valid against the current boundary only; add a margin, report cost and validity under refits; actionability
Overfitting	Explain on held-out data; compare with null importance of noise features
Reason codes	Measure agreement between refits; document variant and baseline; group correlated features
Governance	Explanations are statistics: report method, baseline, sampling error, stability and the question they answer